

Volatility Spillover Effects between Indian Stock Market and Global Stock Markets: A DCC-GARCH Model

FIIB Business Review

1–13

© 2023 Fortune Institute of

International Business

Reprints and permissions:

in.sagepub.com/journals-permissions-india

DOI: 10.1177/23197145221141186

journals.sagepub.com/home/fib

Nikhil Yadav¹, Anurag Bhadur Singh²  and Priyanka Tandon³ 

Abstract

The present article empirically estimates the volatility spillover transmission in Indian equity market represented by Sensex from world economies composite index (Euro Stoxx 50) using the dynamic conditional correlation generalized autoregressive conditional heteroscedasticity (DCC-GARCH) model. The study uses secondary data spanning between 1 April 2012 and March 2022 on weekly basis. The DCC-GARCH model is applied to examine the spillover from developed stock markets to Indian stock market (Sensex). The findings of the study revealed that in short run there is a spillover effect from global markets to Indian stock markets. Investors can invest in the Indian stock market for the long period of time as there is no volatility spillover or volatility transmission from Euro and Nasdaq however in short run the investment in the Indian stock market is not safe due to the presence of volatility effect from all developed stock markets.

Keywords

Volatility spillovers, return, price discovery, ARCH model, DCC-GARCH model and Sensex

Introduction

Spillovers are price changes in one asset that affect the price dynamic behaviour of another. Spillovers have gained prominence in the financial industry due to their diverse applications in portfolio design and risk management, as well as for policymakers in establishing capital market regulations. Financial history is littered with global economic ups and downs, and the recurrence of crises emphasizes the importance of identifying the root cause of financial hardship that leads to the accumulation. Stock market internationalization, liberalized capital flows and massive foreign investment in Indian equity markets have made stock and foreign exchange markets increasingly interdependent (Mishra et al., 2007). Recognizing intermarket volatility is critical for pricing securities within and across markets for trading and hedging strategies, as well as for the formulation of regulatory policies in an emerging market such as India, which is rapidly integrating into the global economy. Global economies have a competent integration in the information-sharing process as well as there exists significant replication of that information

in different financial markets (Joshi, 2011). Investors are especially interested in volatility transmission across international markets because they must constantly monitor and evaluate changes in stock market connections in order to reap the benefits of portfolio diversification and risk-sharing (Jung & Maderitsch, 2014; Kocaarslan et al., 2017). The issue of stock market integration has been focused by the liberalization of the financial markets (Jarungkitkul & Sukcharoensin, 2016). The pace of global integration led to a need to conduct a study to examine the mechanism of stock market dynamics and volatility behaviour.

A reflection of information from one equity market to another equity market generates volatility in return and share prices. This volatility gives new magnitudes in the innovation of price discovery of equity markets. Price discovery is the determination of a share price where buyer and seller implement buying and selling contracts. The Price discovery mechanism of equity in short term can be defined as a process that considers the speed at which the price of equity reacts to new information more specifically in respect of future market price discovery. This indicates the lead-lag relationship between futures and spot prices

¹ Jaypee Business School, Jaypee Institute of Information Technology, Noida, Uttar Pradesh, India

² School of Commerce, XIM University, Bhubaneswar, Odisha, India

³ Regenesys Business School (India Office), Vashi, Mumbai, Maharashtra, India

Corresponding author:

Anurag Bhadur Singh, XIM University, Bhubaneswar, Odisha, India.

E-mail: mailto:anuragsingh.6324@gmail.com

(Booth et al., 1999; Theissen, 2012). Price discovery is the reflection of information into prices, when several markets are correlated, among that information, which information are colloquial material to these markets may be revealed at different speeds of change across these markets. Aitken et al. (2008, p. 21) argued that price discovery is the process by which information is incorporated into stock prices. Investors and regulators are becoming more interested in information transmissions (return and volatility) across stock markets as global financial integration grows. Portfolio managers, for example, must adjust their asset allocations if asset volatility is transferred from one market to another during times of turbulence or crisis (Bouri, 2013; Hung, 2021; Syriopoulos et al., 2015; Vo & Ellis, 2018; Yousaf et al., 2020)

The volatility of the global equity market is divergent and thus leads to fluctuation in major stock market indices of the world. Equity market are much slower than energy market when it comes to the reaction for economic crises (Ashok et al., 2022). Volatility may also lead to a swing in the stock index in any direction either upward or downward. The apprehension behind the volatility is the flow of information in the global economy because positive and negative information affects the sentiments of investors, so global volatility also spills over in the Indian equity market.

This is the reason behind considering this aspect to know the volatility of emerging stock markets like Indian stock market, that is, Sensex because now the Indian economy is an open economy and monitoring the significance of the

global economy on Sensex is an important part of price discovery of the stock market. For financial market players, global volatility helps in the assessment of the risk because variation in share prices is the significant factor in the evaluation risk of equity assets.

Against this backdrop, this article seeks to comprehend differences in equity market reaction between India (Sensex) and leading developed countries (American stock market and Eurozone stock market). The equity indices used for this study are Sensex (India), Nasdaq and New York Stock Exchange (NYSE) (American stock market) and EURO (Euro Stoxx 50).

The volatility of the stock price has mainly been studied in the context of the developed economies (Kaur, 2004). After the most famous study by Engle (1982, p. 989) on the auto-regressive conditional heteroskedasticity (ARCH) model and its generalized form (Generalized Autoregressive Conditional Heteroscedasticity, GARCH) by Bollerslev (1986, p. 309), many studies have used these models to study the volatility in the stock markets as presented in Table 1. There is relatively less study on the emerging economy like India especially with reference to Sensex. In the Indian context, Karmakar (2005, p. 23) focussed on conditional volatility in the Indian stock market. Kaur (2004) exhibits the nature and characteristics of the Indian stock market. Ahmad et al. (2005, p. 197) examine the dynamic linkage between NSE NIFTY and US NASDAQ. Mitra (2017, p. 15) examines the volatility between the Indian Stock market and the foreign exchange market.

Table 1. Key Studies on Volatility Spillover in India over the Years.

Author and Year	Objective	Methodology
Kaur (2004)	Nature of stock market volatility in the Indian stock market	ARCH and GARCH
Karmakar (2005)	Conditional volatility in the Indian stock market	GARCH
Karmakar (2007)	Asymmetric volatility in the Indian stock market	EGARCH
Mishra et al. (2007)	Volatility spillovers between the Indian stock and foreign exchange markets	ARCH and E-GARCH
Mukherjee and Mishra (2010)	Volatility spillovers between India and other Asian Stock markets	GARCH model
Yilmaz (2010)	Volatility spillover in East Asian stock markets	VAR model
Jebran and Iqbal (2016)	Volatility spillover effects between the stock market and foreign exchange market in selected Asian countries; Pakistan, India, Sri Lanka, China, Hong Kong and Japan	EGARCH
Mitra (2017)	Volatility spillovers between the Indian stock and foreign exchange markets	GARCH model
Nandy and Chattopadhyay (2019)	Interdependence between Indian stock and other domestic markets	(DCC-MV-TARCH) (1, 1) model
Sahoo et al. (2019)	Returns spillovers between currency and bond markets	BEKK-GARCH
Uludag and Khurshid (2019)	Volatility spillover from the Chinese stock market to E7 stock market	GARCH
Yadav et al. (2020)	Volatility spillover from equity markets to commodities market in India	Granger causality and dynamic conditional correlation
Dey and Sampath (2020)	Volatility spillovers in three sectors of the Indian stock market	VAR model
Ngene (2021)	Time-varying and directional connectedness of nine equity sectors through intra- and inter-sector volatility spillover in the US	Quantile regression
Oliyide et al. (2021)	Volatility between oil shocks and metal markets	Causality test

Source: Compiled by the authors.

Sakthivel et al. (2012) explored spillover of developed countries index S&P 500(USA), BSE 30 Sensex, FTSE 100, Nikkei 225 and Ordinary Share Price Index but given limited empirical aspect of the European market although in this study, it has been reported European market transmission of information and volatility spillover. Later on Habiba et al. (2020) tried to analyse spillover dynamics among USA and South Asian equity markets and exponential generalized autoregressive conditional heteroskedasticity (EGARCH) model reported spillover of volatility of USA market in South Asian equity markets. Both studies tried to scrutinize the spillover of volatility of Indian equity market and at same time ignorance of European stock market is potential research opportunity for researchers because this index is a representative group of emerging markets and somehow their nature of economy is similar to India, further post Brexit of UK an empirical literature gap exist in the area of global volatility spillover. Euro index and economic area have global importance because it diversified, integrated and share emerging trends like other developing nation further Euro index and economic activities have crucial consequences for the global financial market (Moerman, 2008). For intensive literature survey some of the key studies are presented in Table 1.

The objective of this article is to investigate the following four specific research questions: *first*, is the volatility of one market causing the volatility of another? *Second*, is an asset's volatility transmitted to another asset directly (via its conditional variance) or indirectly (via its conditional covariances)? *Third*, does a shock in one market increase volatility in another, and if so, by how much? *Fourth*, what is the magnitude of spillover in long-run and short-run period to time? Henceforth, the present study contributes in the following ways *first*, to understand the dynamics of the Indian stock market volatility during the COVID-19 pandemic. *Second*, understanding the relationship between the Indian stock market and developed economies. *Third*, the implication of such existing volatility on the investment pattern of the investors. The present study applies the dynamic conditional correlation generalized autoregressive conditional heteroscedasticity (DCC-GARCH) model to examine the spillover effect from the developed economies to the Indian stock market.

Further, the study is organized as follows: The second section presents the literature review, the third section discusses the methodology, the fourth section presents the results and discussion, the fifth section presents concluding remarks, the sixth section discusses the implication to policymakers and investors and the seventh section presents limitations and future scope of research.

Literature Review

Financial market volatility has been extensively studied using the ARCH-GARCH methodology explored by Engle

(1982) and further modified by Bollerslev (1986, p. 315), Nelson (1991) and others. Initially, the research considers the univariate ARCH-GARCH framework to model volatility. Further research marked a withdrawal from univariate volatility models to the multivariate frameworks. These studies generally used the multivariate-GARCH framework to model the conditional variances and covariances across financial markets (Ghosh, 2014). The most widely used model in this class is the vector error correction-generalized autoregressive conditional heteroscedasticity (VEC-GARCH) model (Bollerslev et al., 1988, p. 118) and the Baba-Engle-Kraft-Kroner (BEKK) BEKK model of Engle and Kroner (1995).

Volatility in the Stock Market—International Evidence

The Financial literature depicts the vast number of studies depicting the volatility spillovers across various markets. These regimes cover the recent global financial crisis as well as the protests in Hong Kong. Ross (1989, p. 5) emphasized that apart from price, volatility is also one of the important sources of information for the financial markets. Common factor/news that affects the financial variables mainly affects the first set of volatility spillover (Bollerslev et al., 1992, p. 15). The author states that if the markets are interlinked then variability in common factors is most likely to lead to volatility spillover across markets. The next set of volatility spillovers is due to cross-market hedging (Ederington & Lee, 1993, p. 1165). The financial literature covered vast studies on volatility spillover in stock markets, namely, volatility spillover from one country to another, volatility spillover across indices, volatility spillover from one index to another. Another set of studies covered volatility spillover from spot to futures market and vice-versa. Tse and Booth (1995, p. 5) studied the relationship between US treasury bills and Eurodollar Futures, Tse (1999, p. 915) examined the nexus between the Dow Jones Industrial Average (DJIA) spot and futures market. Ebrahim (2000) used Trivariate GARCH to assess the information dissemination between foreign exchange and money markets in Canada. Ågren (2006) found the volatility spillover from oil prices to stock prices in Japan, Norway, UK and USA. Fedorova and Saleem (2009) found the direct linkage between the equity market (European countries Poland, Hungary, Russia and the Czech Republic) and the currency market. Moon and Yu (2014) examine the short-run volatility spillover effects of daily stock returns and volatilities in the S&P 500 stock index in the US and Shanghai stock exchange in China. The study found symmetric volatility between the markets. There is a strong volatility spillover from US stock markets to other international markets, especially during the recession in the USA (Wang et al., 2017). The study suggested that the information from US markets may help in improving the forecasting accuracy of stock prices. Bissoondoyal-Bheenick

et al. (2018) examined the volatility spillover between the US, China and Australian stock markets and reported the existence of the bilateral existence of the relationship between the three stock markets. Similarly, Caloia et al. (2018) examined the strength of volatility between the five European Economic and Monetary Union (EMU) markets. Alqahtani (2020) analysed the spillover between return and oil prices. In this prospective, Zhang et al. (2020) have conducted research to check the volatility spillover and construct volatility network of G20 countries and found due to economic and trade connection volatility spillover and risk of G20 transmit in USA market. In line with this, there are notable studies conducted on volatility in commodity markets in recent times (Aziz & Hussain, 2021; Cevik et al., 2021; Cui et al., 2021; Maitra et al., 2021). Yasir and Onder (2022) investigated the time-varying herding spillover behaviour in BRIC (Brazil, Russia, India and China) and Turkey under different time horizons. The findings suggest that herding behaviour exists in the Chinese stock market under two different regimes. Ahmed et al. (2022) investigate the dynamics of return linkages and volatility spillovers between Asian emerging stock markets (China, Hong Kong, Japan, Malaysia, Pakistan and South Korea) using bivariate EGARCH(1) model. Results indicated that asymmetric volatility spillovers are also significant in all sampled stock markets except China, according to the findings. With respect to Islamic and conventional stock markets, Bossman et al. (2022) investigated the time and frequency domain approach using time-varying vector autoregressions. The findings show that volatility spillovers across and within Islamic and/or G7 markets vary in time and frequency, but during market turbulence, conventional stocks are more volatile than Islamic stocks.

Volatility in the Stock Markets—Indian Evidence

In comparison to the global studies, there are few findings of Indian markets. Kadapakkam et al. (2003) and Ghosh (2014) studied the Volatility spillover in the foreign exchange market with Indian experience using a multivariate GARCH model author found significant volatility co-movements from different financial markets to the forex market in India. Badrinath and Apte (2005) studied the interlinkage between money, the forex market and the stock market and found significant volatility spillover across the markets. Sarwar (2005) examined the relationship of future volatility of the S&P 500 index and trading volume of options in the S&P 500. For empirical testing, the EGARCH model has been applied and found that the trading volume of option index helps to predict the volatility of future index and have a significant relationship. Mishra et al. (2007) analysed the volatility spillover between the stock market and foreign exchange market in India. The author used a different order autoregressive

generalized autoregressive conditional heteroscedasticity (AR-GARCH) model to estimate the volatility spillover in different markets. The study revealed that there is a bidirectional volatility spillover between Indian stock and forex markets apart from S7P CNX NIFTY and S&P CNX 500. The study also found that both markets move together. Joshi (2011) has investigated volatility in return of Asian economies like India, Hong Kong, Japan, China, Jakarta and Korea, for the period of February 2007 to 29 February 2010. Using GARCH-BEKK, the author found low spillover and integration across markets. Kadapakkam et al. (2003) and Sehgal et al. (2015) examined the relationship between four global currencies and Indian stock exchanges. Using VECM, Johansen Cointegration test and GARCH model, researchers found a long-run equilibrium relationship among future and spot market and futures prices lead to spot prices in the short run. Singhal and Ghosh (2016) in their study investigated the co-movement of crude oil prices and metal prices with BSE Sensex. Results revealed that the volatility of crude oil price and metal does not reflect directly on all Indian stock prices, but some industry has a significant influence. Gahlot (2019) studied the effect of foreign institutional investors (FII) and domestic institutional investor (DII) on the volatility of the stock market. Nandy and Chattopadhyay (2019) examined the volatility between Indian stock markets and other domestic markets such as the foreign exchange market, bullion market and foreign institutions investment. Yadav et al. (2021, p. 3) have investigated the dynamic relationship of crude oil prices and Sensex, as a result, reported that crude oil prices have only a short-term effect on Sensex. Mishra et al. (2022) use the multivariate GARCH-BEKK model to examine the return and volatility spillover between the Indian stock market and Asian markets. Results clearly confirm that there is volatility transmission from developed economies to Indian market. Yadav et al. (2022) examined the spillover between financial and energy exchange-traded funds. Mahalashmi et al. (2022) investigated the connectedness between the Indian rupee against the US dollar and the stock market. The study confirmed the strong association between the currency value of a stable economy and with the economy of trade. Kumar and Singh (2022) studied the spillover dynamics of crude oil with financial indicators post-WPI revision. The results show an intermittent increase in spillover during the Covid era, the GFC and the oil crisis. Nagarkatte and Natchimuthu (2022) studied the impact of the Brexit referendum on the return and volatility spillover between the EU, the UK and US stock markets and Indian stock markets during pre- and post-Brexit referendum. According to the study's findings, prior to the Brexit referendum, Indian stock market returns had no significant return spillover on other markets. On the contrary, following the referendum, Indian stock returns significantly impacted stock market returns in France, Germany, the United Kingdom and the United States.

Research Gap

With the extensive literature survey, it has been observed that most of the studies have been conducted on individual markets, namely, commodities, currencies, forex, and markets in India in comparison of volatility with different equity markets. However, there have been few studies that focussed on the spillover volatility effect between the global economy and the Indian economy. Hence, this motivates us to conduct a study on volatility with top global markets and their significance on the Indian equity market. The empirical results of the present study will help in adding inferences of global market volatility on the Indian equity market (Sensex). Further, the results will be helpful in identifying the relevant global markets having contagion effects concerning volatility in the Indian stock market. As the perspective of past studies was confined to individual variables only, therefore the present study will add different prospects in the literature.

Data and Methodology

We examine the spillover effect from global stock markets to the Indian stock market. The BSE Sensex is the proxy used for the Indian stock market, whereas global stock markets is represented by three stock markets, NYSE and NASDAQ and Euro Stoxx 50 proxied as EURO. These variables' observations have spanned from April 2012 to March 2022. In this study, the researcher attempted to look at statistical evidence that had spilled over from the post-recession period to the new normal scenario with major global economic players. Governments had implemented a number of prudent economic measures to address various issues, and as a result, the economy had normalized and had gained some financial stability by the time, hence we chose the data set for the study—from 2012 to 2022. The daily adjusted closing prices of all stocks and commodities have been transformed into daily log returns, which are the logarithmic differences in prices over two consecutive days, that is, post 2008 recession shock global economic giants had preferred expansionary budget to move out from existing slowdown of economy but expansionary budget policies were not uniform, which further created disequilibrium in performance of global equity market (Singh & Singh, 2016). Recent pandemic has also down spare information spillover among global markets (Zhao et al., 2022) because of international financial market risk and uneven development of economies (Ren et al., 2022).

$$R_{it} = \log\left(\frac{P_{it}}{P_{it-1}}\right) \quad (1)$$

where R_{it} is the log return at time t and P_{it} and P_{it-1} are the two successive daily closing prices.

The augmented Dickey–Fuller (ADF) test has been used to check the stationarity of the data. Further, DCC has been used to investigate the volatility spillover effect from the global stock market to the Indian stock market.

Stationarity/Unit Root Test

Before using the cointegration approach, it is worth testing whether time-series data are integrated of the same order or not, as well as whether the data is stationary or not, as otherwise the results obtained will be erroneous (Samsu et al., 2008; Yadav et al., 2020). A variety of tests are available to determine the presence of unit root in data, including the Dickey and Fuller test and the tau test (Dickey & Fuller, 1979), the Phillips–Perron test and ADF test (Phillips & Perron, 1988). As per Gujarati & Porter (2017), while conducting the DF test, it was assumed that error terms are uncorrelated. However, if in case error terms are correlated, then Dickey and Fuller have developed another test known as the ADF test. Furthermore, Phillips and Perron employ nonparametric statistical methods to account for serial correlation in error terms without the addition of lagged difference terms. ADF is by far the most widely used method for determining time-series data stationarity (Singh & Sharma, 2018). Mathematically, the equation for ADF can be represented as follows:

$$\Delta Y_t = \beta_1 + \beta_2 + \beta_3 Y_{t-1} + \sum_{i=1}^{k-1} \alpha_i Y_{t-i} + \varepsilon_t \quad (2)$$

The data's non-stationarity (unit root) can be examined in the following three stages: intercept (constant), intercept and trend, and none. For Equation 1, the null hypothesis can be, $\beta_3 = 0$, where β_3 is the coefficient for lagged term Y_{t-1} . The failure to reject the null hypothesis implies that the time series under consideration is nonstationary. A series denoted by $I(d)$ is said to be integrated of order d , if it has to be differenced d times to make it stationary. Henceforth, stationary time-series or time-series integrated of order zero are the same. Therefore, the ADF proposes to test the following hypothesis:

$$\beta_3 = (\text{All series contains the unit root})$$

Dynamic Conditional Correlation (DCC) Model

The literature on spillover provides evidence on the application of various multivariate volatility models that investigate conditional covariance (Yadav et al., 2022). Multivariate models are powerful and flexible applications that provide improved financial decision making, such as asset pricing, portfolio optimization, hedging and risk management (Yadav et al., 2020). The diagonal model (Bollerslev et al., 1988); the diagonal VECH model and multivariate GARCH model (Engle & Kroner, 1995); the vector autoregressive moving-average generalized autoregressive conditional heteroscedasticity (ARMA-GARCH) or vector autoregressive moving-average generalized autoregressive conditional heteroscedasticity (VARMA-GARCH) model (Ling & McAleer, 2003); the DCC-GARCH model (Engle, 2002); and the varying conditional correlation model are notable examples (Tse &

Tsui, 2002). DCC is an appropriate multivariate instrument for the transmission of information from one market to another. Based on a review of these models and their potential explanatory power, the DCC-GARCH model, which was propounded by Engle (2002) and which evaluates time-varying correlations, was used. DCC model examines the time-varying correlation between the series and helps in integration (Yadav et al., 2020). Its estimation entails the following two steps: *first*, application of univariate GARCH. To model R_{it} , the following equation is estimated:

$$R_{it} = a + b_1 R_{i,t-1} + \varepsilon_{it} = h_{it}(\frac{1}{2}) v_{it} \quad (3)$$

where a is constant, b_1 is the coefficient of lagged return, ε_{it} is the random error term that has conditional variance h_{it} while v_{it} is the vector $n \times m$ of residuals that are identically dispersed and independent. In *second* step of DCC-GARCH, correlations are projected using the following equations:

$$C_t = D_t R_t D_t \quad (4)$$

where C_t is the covariance matrix, R_t is the conditional covariance matrix and D_t is the $n \times n$ diagonal matrix with time-varying standard deviations with diagonal.

$$D_t = \text{Diag}(h_1 t/2, \dots, h_n t/2) \quad (5)$$

$$R_t = Q_t^{*-1} Q_t Q_t^{*-1} \quad (6)$$

where Q_t is the symmetric positive definite index

$$Q_t = (1 - a - b) Q' + a \varepsilon_{t-1} \varepsilon'_{t-1} + b Q_{t-1} \quad (7)$$

Q' is the standardized errors' unconditional covariance matrix and Q_t^* is the diagonal matrix made up of the square root of the diagonal of Q_t , which can be represented as $\text{diag}(q_1/211t, q_1/222t, \dots, q_1/2mnt)$. The equation contains two DCC parameters, a and b , both of which are non-negative and have a sum less than one. Lower conditional correlation values indicate greater diversification opportunities, whereas higher values indicate greater integration (Yu et al., 2010). The DCC-GARCH alpha and beta estimators are time-varying. Alpha quantifies the volatility impact over a shorter time period while taking

into account the impact of residual persistence from the previous period. In DCC, beta measures the long-term effect of a shock on conditional correlation. The dynamic correlations are calculated as follows:

$$p_{ij,t} = \frac{q_{ij,t}}{\sqrt{q_{ii,t} q_{jj,t}}} \quad (8)$$

Engle (2002) employs the two-step likelihood estimation method to estimate the DCC-GARCH model. The likelihood function is shown below:

$$Ln(L(\theta)) = -1/2 \sum_{t=1}^T \{n \ln(2\pi) + \ln|D_t|^2 + \ln(|R_t|) + \varepsilon'_t D_t^{-2} \varepsilon_t\} \quad (9)$$

As a result, this is a modelling approach with a time-varying mean, variances and covariance.

Results and Discussion

Table 2 presents the summary statistics for the series used in the study. The log return of all the financial markets realized the positive average returns. Further, the volatility of Nasdaq represented by standard deviation is high (0.46) followed by Sensex (0.31), NYSE (0.19) and the volatility in Euro is less (0.14). The standard deviation value clearly depicts that Indian stock market Sensex is highly volatile as compared to others. The Euro market is negatively skewed while returns of rest of the financial markets are positively skewed. The kurtosis value in case of Nasdaq, NYSE and Sensex is less than -1 , which indicates that the distribution is too flat.

The Jarque-Bera test is applied to test the normality in the series. The statistics derived from test indicates that the series are not normally distributed. Additionally, it is important to test the presence of stationarity in the time-series data. Table 4 presents the results for ADF test. The null hypothesis for the test is: H_0 : *series has unit root*. Since the p -value for the all the series are insignificant at level however they all are significant at first differenced, leading to rejection of null hypothesis at first differenced. Hence, it can be inferred here that all the return series are integrated of order 1, that is, $I(1)$.

Next, we applied the DCC-GARCH to examine the spillover from global financial markets to Indian stock

Table 2. Data Description.

Market	Acronym	Source
Indian Stock Market	SENSEX	www.bseindia.com
American Stock Market	NYSE	www.yahoofinance.com
American Stock Market	NASDAQ	www.yahoofinance.com
Eurozone stock market	EURO	www.yahoofinance.com

Source: The authors.

Table 3. Summary Statistics of Indian and Global Stock Markets.

Euro		Nasdaq		NYSE		Sensex	
Mean	8.0858	Mean	8.7380	Mean	9.3542	Mean	10.3032
Standard error	.0062	Standard error	0.0205	Standard error	0.0084	Standard error	0.0139
Median	8.0992	Median	8.6809	Median	9.3290	Median	10.2628
Standard deviation	0.1422	Standard deviation	0.4686	Standard deviation	0.1919	Standard deviation	0.3179
Sample variance	0.0202	Sample variance	0.2196	Sample variance	0.03682	Sample variance	0.1010
Kurtosis	0.3582	Kurtosis	−0.7806	Kurtosis	−0.2305	Kurtosis	−0.5437
Skewness	−0.5058	Skewness	0.3087	Skewness	0.1224	Skewness	0.1764
Range	0.7479	Range	1.7654	Range	0.8637	Range	1.3514
Minimum	7.6346	Minimum	7.9184	Minimum	8.8945	Minimum	9.6796
Maximum	8.3825	Maximum	9.6839	Maximum	9.7583	Maximum	11.0311
Jarque–Bera statistics							
Euro			71.07				
Nasdaq			31.84				
NYSE			94.34				
Sensex			0.43				
Count	522	Count	522	Count	522	Count	522

Source: The authors.

Table 4. Augmented Dickey–Fuller Test.

Indices	p-Value (at Level)	p-Value (at First Differenced)
Euro	.0692	.01*
Nasdaq	.7983	.01*
NYSE	.3424	.01*
Sensex	.7281	.01*

Source: The authors.

Note: *Significant at 5% level of significance.

market. Table 5 presents the results for the DCC-GARCH model for the current study. The results are segregated into panels. Panel (5A) presents the spillover effect from euro to Sensex, panel (5B) presents the spillover effect from Nasdaq to Sensex and panel (5C) presents the spillover effect from NYSE to Sensex. The DCC-alpha (DCCa1) signifies the spillover effect in the short run, whereas the DCC-beta (DCCb1) signifies the spillover effect in long run. The dccal value is insignificant across all panels in the study, which clearly indicates that there is a presence of no volatility spillover or volatility transmission from global financial markets to the Indian stock market proxied by Sensex in short-run. However, dccb1 parameter for Panel (5A) is insignificant (0.0000), which indicates that there is volatility transmission from reuro to Sensex, that is, Eurozone financial markets to Indian stock market in long-run period. The findings of the study postulate that investor can invest in the Indian stock market for short-run period as there is no sign of volatility spillover from the global stock market considered in the study. But, at the same time, investors need to be very cautious while investing for long-run period as there is an evidence of volatility spillover from Eurozone stock market to Indian stock market.

Additionally, in Panel (5B) and Panel (5C), the dccb1 value is (0.0813) and (0.0000), which is significant at 10% and 5% level of significance, respectively. It indicates that,

over time, there is a spillover or transfer of volatility from the Nasdaq and NYSE to the Sensex in long run. Furthermore, Figure 1 presents the conditional correlation plot of the series. The results give a clear indication that investors can invest in Indian stock market for the short period of time as there is no volatility spillover or volatility transmission from global financial markets as confirmed by the results above however in long-run the investment in Indian stock market is not safe due to the presence of volatility transmission from all developed stock markets. Figure 1(a), 1(b) and 1(c) presents the conditional correlation plot of reuro-sensex, nasdaq-sensex and nyse-sensex, respectively.

As the present study examined the spillover effect between global stock markets represented by Nasdaq, NYSE and Euro Stoxx 50 and Indian stock market proxied by S&P Sensex. The stock market has undergone significant and striking change and due to which it is imperative to study the volatility in the Indian stock market post-recession period (Yadav et al., 2022). After summary statistics, we employed the DCC-GARCH model to examine the volatility spillover among the global and Indian stock markets (Table 3). The results from the DCC-GARCH model revealed that there is the presence of no short-run volatility transmission to the Indian stock market (Sensex) from the global stock market in short run, whereas there is the presence of volatility spillover from Eurozone financial markets to the Indian stock market (Sensex) in long run. With this backdrop, globalization is to blame for the influence of volatility on the Sensex because every nation is now interconnected for economic purposes. As a result, any economic or non-economic developments induce volatility in all emerging equity markets. As the Indian equity market is an emerging equity market and the nature of the market is semi-efficient, it easily comes under the influence of the global equity market. On the other

Table 5. Results of DCC-GARCH Model.

Panel (5A): DCC from reuro to rsensex				
	Estimate	Std. Error	t-Value	Pr(> t)
[rnasdaq].mu	4.4324	3.1642	1.4008	0.1612
[rnasdaq].omega	565.3375	310.1200	1.8229	0.0683***
[rnasdaq].alpha I	0.1604	0.0801	2.0006	0.0454**
[rnasdaq].beta I	0.7738	0.0889	8.6961	0.0000*
[rsensex].mu	75.2063	30.9650	2.4287	0.0151**
[rsensex].omega	597.1252	2371.4	0.2518	0.8011
[rsensex].alpha I	0.0906	0.0261	3.4651	0.0005*
[rsensex].beta I	0.9083	0.0330	27.4476	0.0000*
[Joint]dccal	0.0000	0.0001	0.0000	0.9999
[Joint]dccbl	0.9170	0.2648	3.4623	0.0005*
Panel (5B): DCC from rnasdaq to rsensex				
	Estimate	Std. error	t-value	Pr(> t)
[rnasdaq].mu	18.3857	3.9648	4.6372	0.0000*
[rnasdaq].omega	491.5371	197.25	2.4919	0.0127**
[rnasdaq].alpha I	0.2601	0.0453	5.7384	0.0000*
[rnasdaq].beta I	0.7388	0.0372	19.8316	0.0000*
[rsensex].mu	75.2063	30.950	2.4298	0.0151**
[rsensex].omega	597.1252	2360.8	0.2529	0.8003
[rsensex].alpha I	0.0906	0.0261	3.4675	0.0005*
[rsensex].beta I	0.9083	0.0329	27.5271	0.0000*
[Joint]dccal	0.0000	0.0000	0.0102	0.9918
[Joint]dccbl	0.9061	0.5198	1.7431	0.0813***
Panel (5C): DCC from rNYSE to rsensex				
	Estimate	Std. Error	t-value	Pr(> t)
[rnyse].mu	1.9316e+01	7.2556	2.6622	0.0077*
[rnyse].omega	2.3116e+03	948.89	2.4361	0.0148**
[rnyse].alpha I	2.5113e-01	748.80	3.3537	0.0007*
[rnyse].beta I	7.4787e-01	365.34	20.4703	0.0000*
[rsensex].mu	7.5206e+01	30.996	2.4263	0.0152**
[rsensex].omega	5.9713e+02	2403.3	0.2484	0.8037
[rsensex].alpha I	0.647e-02	262.89	3.4481	0.0005**
[rsensex].beta I	9.0835e-01	334.64	27.1445	0.0000**
[Joint]dccal	1.3111e-02	281.57	0.4656	0.6414
[Joint]dccbl	8.8345e-01	406.51	21.7328	0.0000*

Source: The authors.

Note: DCC stands for dynamic conditional correlation, reuro stands for return on eurozone stock market, rsensex is return on Indian stock market (Sensex), rnasdaq is return on NASDAQ, rNYSE is return on New York stock exchange, dccal signifies the dcc-alpha I, which is spillover in short-run, dccbl signifies the dcc-beta I, which is spillover in long-run.

*Significant at 1% significance level, **significant at 5% significance level and ***significant at 10% level.

hand, international portfolio diversification by investors also leads to volatility in semi-efficient markets for a very short-run period. Similar findings have been confirmed by (Sakthivel et al., 2012) where they found bidirectional volatility transmission from US stock markets to Sensex in the long-run and such integration is reported due to strong trade and investment connections. Interestingly, Habiba et al. (2020) confirmed the volatility transmission from US stock markets to Indian stock market in long-run in post-crisis period. On similar note, Trivedi and Birau (2020) revealed that there has been high volatility in Indian stock market after global financial crisis. Additionally, Spullber et al. (2021) also confirmed that there has been volatility transmission from European stock markets to Indian stock market.

Kadapakka et al. (2003, p. 170) also confirm the same results as they investigated the role of the foreign equity market in the price discovery of Indian equity markets, in their study London global depository receipts market and London equity market is taken as representative of the foreign equity market where researchers found in the study, foreign equity market spillover in price discovery of the Indian equity market. Mukherjee and Mishra (2010, p. 236) also found that major Asian countries' equity markets volatility spillover on the Indian equity market and Indian equity market incorporate the information in the price discovery process without delay.

Most of the previous literature on the volatility of the equity market was related to exchange rate, commodity and equity return, hence the findings of this study will add

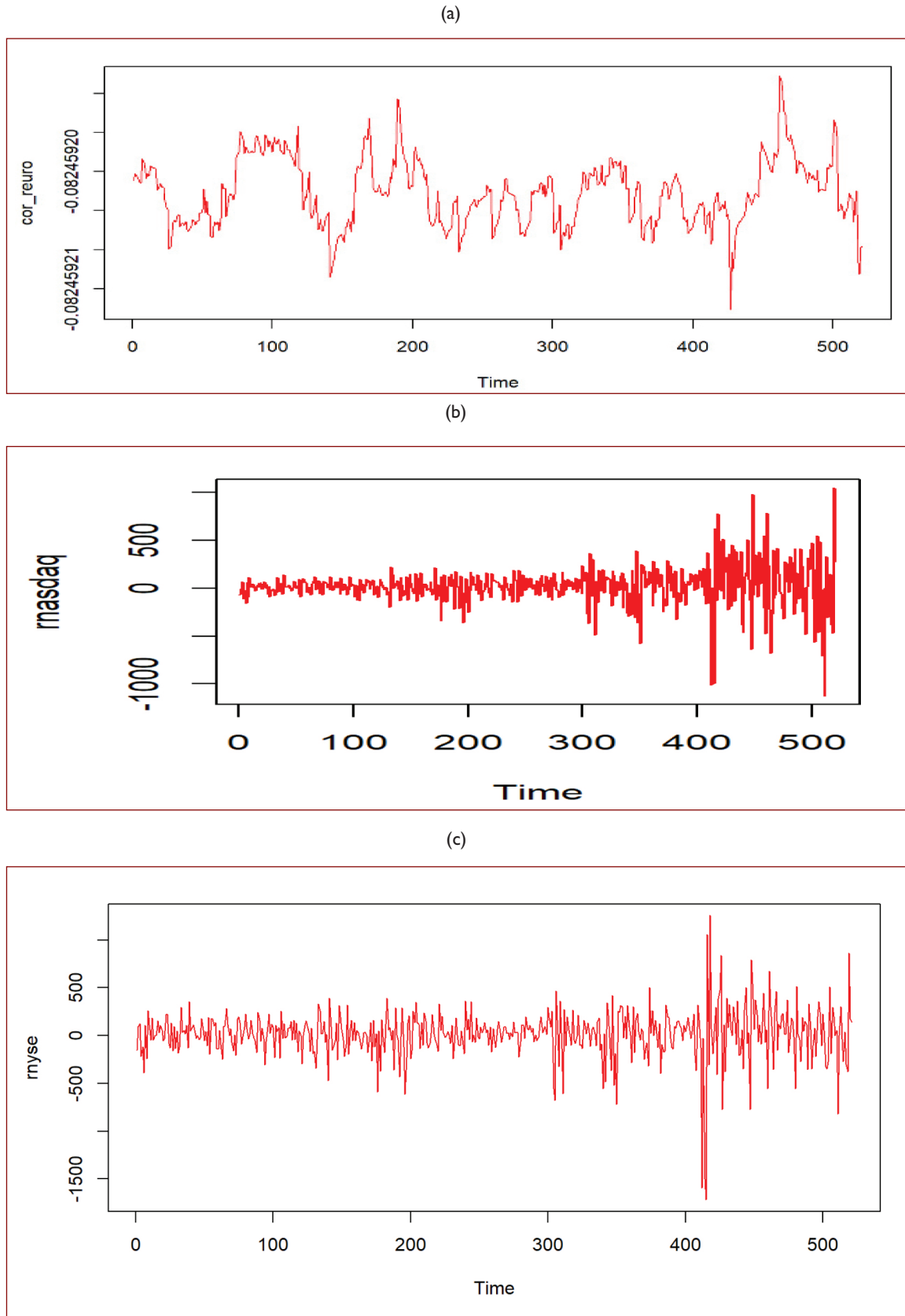


Figure 1. Time-Varying Correlation among Constituent Series. (a) DCC Conditional Correlation reuro-rsensex. (b) DCC Conditional Correlation rnasdaq-rsensex. (c) DCC Conditional Correlation rnyse-rsensex.

Source: The authors.

to the available literature on the volatility of the equity market because the dimension of the study was confined to Sensex and major emerging global equity market's volatility.

Concluding Remarks

The article investigates the volatility spillover from different global markets to the Indian stock market using weekly data from the year 2013 to 2018. GARCH methodology is extensively applied to capture the Stoox 50 index weekly returns and volatility, which not only affects the conditional returns but also affects the conditional volatility weekly. Research considers the univariate ARCH-GARCH framework to model volatility. Further research marked a withdrawal from univariate volatility models to the multivariate framework. These studies generally used the multivariate-GARCH framework to model the conditional variances and covariances across financial markets. The study employed the DCC-GARCH model to overcome the shortcomings of the ARCH model. The findings of the study revealed that in short run there is spillover effect from global markets to Indian stock markets. Investors can invest in Indian stock market for the long period of time as there is no volatility spillover or volatility transmission from Euro and Nasdaq however in short run the investment in Indian stock market is not safe due to the presence of volatility effect from all developed stock markets. Further investor can make proper portfolio diversification in light of developed countries stock movement.

Implications of the Study

Implications to Policymakers

To encourage growth through established financial markets while minimizing the danger to financial stability. In order to maintain financial stability worldwide and to understand the effects of spillover or contagion, policymakers are interested in market co-movement. Policymakers and financial regulators can take some corrective action in advance to protect the investor interest and level of markets. If any country has spillover connection, then policymakers then they can take a few short-term steps to ensure market stability. Other side with global market co-movement and spillover information, investor and portfolio manager can diversify their portfolio in different equity market with better asset allocation strategy. The findings of this study will help to ascertain economic impact of globalization and regional integration on global equity markets because in era of diversification switch in different investment regime have a pivotal role of portfolio diversification that further switch in different investment regime help global investor to avoid country specifics or region specific economic consequences.

Implications to Investors

The results of the study have significant contributions and implications for the investors as well. The investors must perform a post-analysis of each investment so that they can be aware of the past behaviour of stock market dynamics. Both internal and external events contribute to changing dynamics of both return and volatility spillovers (Dey & Sampath, 2020). Therefore, investors need to be careful about volatility-based trading strategies. Investors should have a constant watch on the behaviour and investment pattern of FIIs especially at the time of crisis when the outflow is more than inflow. It is important that fund managers need to identify the global investment behaviour of investors and likely behavioural biases before designing the portfolio.

Limitation and Future Scope of Research

The study considered limited stock indices outside the Indian equity market and performed for a limited time period. Further, the role of black swan events such as the COVID-19 pandemic and some major systematic risk factors may be considered for measuring the volatility in the Indian stock market.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The authors received no financial support for the research, authorship and/or publication of this article.

ORCID iDs

Anurag Bhadur Singh  <https://orcid.org/0000-0003-3363-4686>
Priyanka Tandon  <https://orcid.org/0000-0002-8077-0317>

References

- Ågren, M. (2006). Does oil price uncertainty transmit to stock markets? <http://www.nek.uu.se>
- Ahmad, K. M., Ashraf, S., & Ahmed, S. (2005). Is the Indian stock market integrated with the US and Japanese markets? An empirical analysis. *South Asia Economic Journal*, 6(2), 193–206.
- Ahmed, R. I., Zhao, G., & Habiba, U. (2022). Dynamics of return linkages and asymmetric volatility spillovers among Asian emerging stock markets. *The Chinese Economy*, 55(2), 156–167. <https://doi.org/10.1080/10971475.2021.1930292>
- Aitken, M., Almeida, N., Harris, F. H. D., & McNish, T. H. (2008). Financial analysts and price discovery. *Accounting & Finance*, 48(1), 1–24.
- Alqahtani, A., & Chevallier, J. (2020). Dynamic spillovers between Gulf Cooperation Council's stocks, VIX, oil

- and gold volatility indices. *Journal of Risk and Financial Management*, 13(4), 69.
- Ashok, S., Corbet, S., Dhangra, D., Goodell, J. W., Kumar, S., & Yadav, M. P. (2022). Are energy markets informationally smarter than equity markets? Evidence from the COVID-19 experience. *Finance Research Letters*, 47(B), 102728. <https://doi.org/10.1016/j.frl.2022.102728>
- Aziz, T., & Hussain, A. (2021). Volatility spillovers of gold prices, oil prices, and economic policy uncertainty on the stock market of Pakistan. *Global Business and Economics Review*, 24(4), 344–359.
- Badrinath, H. R., & Apte, P. G. (2005). *Volatility spillovers across stock, call money and foreign exchange markets*. Department of Economics (San Diego, University of California). https://www.researchgate.net/publication/271765360_Volatility_spillovers_across_stock_call_money_and_foreign_exchange_markets
- Bissoondoyal-Bheenick, E., Brooks, R., Chi, W., & Do, H. X. (2018). Volatility spillover between the US, Chinese and Australian stock markets. *Australian Journal of Management*, 43(2), 263–285.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Bollerslev, T., Chou, R. Y., & Kroner, K. F. (1992). ARCH modelling in finance: A review of the theory and empirical evidence. *Journal of Econometrics*, 52(1–2), 5–59.
- Bollerslev, T., Engle, R. F., & Woodredge, J. M. (1988). A capital asset pricing model with time-varying covariances. *Journal of Political Economy*, 96(1), 116–131.
- Booth, G. G., So, R. W., & Tse, Y. (1999). Price discovery in the German equity index derivatives markets. *Journal of Futures Markets: Futures, Options, and Other Derivative Products*, 19(6), 619–643.
- Bossmann, A., Junior, P. O., & Tiwari, A. K. (2022). Dynamic connectedness and spillovers between Islamic and conventional stock markets: Time-and frequency-domain approach in COVID-19 era. *Heliyon*, 8(4), e09215. <https://doi.org/10.1016/j.heliyon.2022.e09215>
- Bouri, E. I. (2013). Correlation and volatility of the MENA equity markets in turbulent periods, and portfolio implications. *Economics Bulletin*, 33(2), 1575–1593.
- Caloia, F. G., Cipollini, A., & Muzzioli, S. (2018). Asymmetric semi-volatility spillover effects in EMU stock markets. *International Review of Financial Analysis*, 57, 221–230. <https://doi.org/10.1016/j.irfa.2018.03.001>
- Cevik, E. I., Dibooglu, S., Abdallah, A. A., & Al-Eisa, E. A. (2021). Oil prices, stock market returns, and volatility spillovers: Evidence from Saudi Arabia. *International Economics and Economic Policy*, 18(1), 157–175.
- Cui, J., Goh, M., & Zou, H. (2021). Coherence, extreme risk spillovers, and dynamic linkages between oil and China's commodity futures markets. *Energy*, 225, 120190.
- Dey, S., & Sampath, A. (2020). Returns, volatility and spillover—A paradigm shift in India? *The North American Journal of Economics and Finance*, 52, 101–110. <https://doi.org/10.1016/j.najef.2019.101110>
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366a), 427–431.
- Ebrahim, S. K. (2000). *Volatility transmission between foreign exchange and money markets* (Bank of Canada Working Paper 2000-16). bankofcanada.ca/2000/08/working-paper-2000-16/
- Ederington, L. H., & Lee, J. H. (1993). How markets process information: News releases and volatility. *The Journal of Finance*, 48(4), 1161–1191.
- Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance. *Econometrica*, 50(4), 987–1007.
- Engle, R. F., & Kroner, K. F. (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory*, 11(1), 122–150.
- Fedorova, E., & Saleem, K. (2009, August). *Volatility spillovers between stock and currency markets: Evidence from emerging Eastern Europe* [Paper presentation]. 22nd Australasian Finance and Banking Conference. <https://ssrn.com/abstract=1460645>
- Gahlot, R. (2019). An analytical study on effect of FIIs & DIIs on Indian stock market. *Journal of Transnational Management*, 24(2), 67–82.
- Ghosh, S. (2014). Volatility spillover in the foreign exchange market: The Indian Experience. *Macroeconomics and Finance in Emerging Market Economies*, 7(1), 175–194.
- Gujarati, D. N., & Porter, D. C. (2017). *Basic econometrics* (sixth ed.). McGraw-Hill/Irwin.
- Habiba, U. E., Peilong, S., Zhang, W., & Hamid, K. (2020). International stock markets integration and dynamics of volatility spillover between the USA and South Asian markets: Evidence from Global financial crisis. *Journal of Asia Business Studies*, 14(5), 779–794.
- Hung, N. T. (2021). Volatility behaviour of the foreign exchange rate and transmission among central and eastern European countries: Evidence from EGARCH model. *Global Business Review*, 22(1), 36–56.
- Jarungkitkul, W., & Sukcharoensin, S. (2016). Benchmarking the competitiveness of the ASEAN 5 equity markets: An application of Porter's diamond model. *Benchmarking: An International Journal*, 23(5), 1312–1340.
- Jebran, K., & Iqbal, A. (2016). Dynamics of volatility spillover between stock market and foreign exchange market: Evidence from Asian Countries. *Financial Innovation*, 2(1), 1–20.
- Joshi, P. (2011). Return and volatility spill-overs among Asian stock markets. *SAGE Open*, 1(1). <https://doi.org/10.1177/2158244011413474>
- Jung, R. C., & Maderitsch, R. (2014). Structural breaks in volatility spillovers between international financial markets: Contagion or mere interdependence? *Journal of Banking & Finance*, 47, 331–342.
- Kadapakkam, P. L., Misra, L., & Tse, Y. (2003). International price discovery for emerging market stocks: Evidence from Indian GDRs. *Review of Quantitative Finance and Accounting*, 21(2), 179–199.
- Karmakar, M. (2005). Modeling conditional volatility of the Indian stock markets. *Vikalpa*, 30(3), 21–38.
- Karmakar, M. (2007). Asymmetric volatility and risk-return relationship in the Indian stock market. *South Asia Economic Journal*, 8(1), 99–116.

- Kaur, H. (2004). Time varying volatility in the Indian stock market. *Vikalpa*, 29(4), 25–42.
- Kocaarslan, B., Sari, R., Gormus, A., & Soytaş, U. (2017). Dynamic correlations between BRIC and US stock markets: The asymmetric impact of volatility expectations in oil, gold and financial markets. *Journal of Commodity Markets*, 7, 41–56.
- Kumar, P., & Singh, V. K. (2022). Examining the time varying spillover dynamics of indian financial indicators from global and local economic uncertainty. *Journal of Quantitative Economics*, 20(4). <https://doi.org/10.1007/s40953-022-00333-8>
- Ling, S., & McAleer, M. (2003). Asymptotic theory for a vector ARMA-GARCH model. *Econometric Theory*, 19(2), 280–310.
- Mahalakshmi, S., Thiagarajan, S., Vasudevan, G., & Naresh, G. (2022). Return and volatility spillover effects between rupee-dollar exchange rate and asian stock indices. *Journal of Emerging Market Finance*, 21(4). <https://doi.org/10.1177/09726527221100467>
- Maitra, D., Guhathakurta, K., & Kang, S. H. (2021). The good, the bad and the ugly relation between oil and commodities: An analysis of asymmetric volatility connectedness and portfolio implications. *Energy Economics*, 94, 105061. <https://doi.org/10.1016/j.eneco.2020.105061>
- Mishra, A. K., Agrawal, S., & Patwa, J. A. (2022). Return and volatility spillover between India and leading Asian and global equity markets: an empirical analysis. *Journal of Economics, Finance and Administrative Science*, 27(54). pp. 294–312. <https://doi.org/10.1108/JEFAS-06-2021-0082>
- Mishra, A. K., Swain, N., & Malhotra, D. K. (2007). Volatility spillover between stock and foreign exchange markets: Indian evidence. *International Journal of Business*, 12(3), 343–359.
- Mitra, P. K. (2017). Dynamics of volatility spillover between the Indian stock market and foreign exchange market return. *Academy of Accounting and Financial Studies Journal*, 21(2), 1–11.
- Moerman, G. A. (2008). Diversification in euro area stock markets: Country versus industry. *Journal of International Money and Finance*, 27(7), 1122–1134. <https://doi.org/10.1016/j.jimonfin.2008.05.005>
- Moon, G. H., & Yu, W. C. (2010). Volatility spillovers between the US and China stock markets: structural break test with symmetric and asymmetric GARCH approaches. *Global Business Review*, 39(2), 129–149
- Mukherjee, K., & Mishra, R. K. (2010). Stock market integration and volatility spillover: India and its major Asian counterparts. *Research in International Business and Finance*, 24(2), 235–251.
- Nagarakatte, S. G., & Natchimuthu, N. (2022). Return and volatility spillover between India, UK, USA and European stock markets: The Brexit impact. *Investment Management and Financial Innovations*, 19(1), 121–134. doi:10.21511/imfi.19(1).2022.09
- Nandy, S., & Chattopadhyay, A. F. (2019). Indian stock market volatility: A study of inter-linkages and spill-over effects. *Journal of Emerging Markets and Finance*, 18(2_suppl), S183–S212.
- Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370.
- Ngene, G. M. (2021). What drives dynamic connectedness of the US equity sectors during different business cycles? *The North American Journal of Economics and Finance*, 58, 101493. <https://doi.org/10.1016/j.najef.2021.101493>
- Oliyide, J. A., Adekoya, O. B., & Khan, M. A. (2021). Economic policy uncertainty and the volatility connectedness between oil shocks and metal market: An extension. *International Economics*, 167, 136–150. <https://doi.org/10.1016/j.inteco.2021.06.007>
- Phillips, P. C., & Perron, P. (1988). Testing for a unit root in time series regression. *Biometrika*, 75(2), 335–346.
- Ren, Y., Tan, A., Zhu, H., & Zhao, W. (2022). Does economic policy uncertainty drive nonlinear risk spillover in the commodity futures market? *International Review of Financial Analysis*, 81, 102084. <https://doi.org/10.1016/j.irfa.2022.102084>
- Ross, S. A. (1989). Information and volatility: The no-arbitrage martingale approach to timing and resolution irrelevancy. *The Journal of Finance*, 44(1), 1–17.
- Sahoo, S., Behera, H., & Trivedi, P. (2019). Return and volatility spillovers between currency and bond markets in India. *Macroeconomics and Finance in Emerging Market Economies*, 12(2), 155–173.
- Sakthivel, P., Bodkhe, N., & Kamaiah, B. (2012). Correlation and volatility transmission across international stock markets: A bivariate GARCH analysis. *International Journal of Economics and Finance*, 4(3), 253–264.
- Samsu, S. H., Derus, A. M., Ooi, A. Y., & Ghazali, M. F. (2008). Causal links between foreign direct investment and exports: Evidence from Malaysia. *International Journal of Business and Management*, 3(12), 177–183.
- Sarwar, G. (2005). The informational role of option trading volume in equity index options markets. *Review of Quantitative Finance and Accounting*, 24(2), 159–176.
- Sehgal, S., Ahmad, W., & Deisting, F. (2015). An investigation of price discovery and volatility spillovers in India's foreign exchange market. *Journal of Economic Studies*, 42(2), 261–284.
- Singh, A., & Singh, M. (2016). Cross country co-movement in equity markets after the US financial crisis: India and major economic giants. *Journal of Indian Business Research*, 8(2), 98–121.
- Singh, N. P., & Sharma, S. (2018). Phase-wise analysis of dynamic relationship among gold, crude oil, US dollar and stock market. *Journal of Advances in Management Research*, 15(4), 480–499. <https://doi.org/10.1108/JAMR-12-2017-0124>
- Singhal, S., & Ghosh, S. (2016). Returns and volatility linkages between international crude oil price, metal and other stock indices in India: Evidence from VAR-DCC-GARCH models. *Resources Policy*, 50, 276–288. <https://doi.org/10.1016/j.resourpol.2016.10.001>
- Theissen, E. (2012). Price discovery in spot and futures markets: A reconsideration. *The European Journal of Finance*, 18(10), 969–987.
- Tse, Y. (1999). Price discovery and volatility spillovers in the DJIA index and futures markets. *Journal of Futures Markets*, 19(8), 911–930.
- Tse, Y., & Booth, G. G. (1995). The relationship between U.S. and Eurodollar interest rates: Evidence from the futures

- markets. *Review of World Economics*, 131(1), 28–46. Tse, Y. K., & Tsui, A. K. C. (2002). A multivariate generalized autoregressive conditional heteroscedasticity model with time-varying correlations. *Journal of Business & Economic Statistics*, 20(3), 351–362.
- Uludag, B. K., & Khurshid, M. (2019). Volatility spillover from the Chinese stock market to E7 and G7 stock markets. *Journal of Economic Studies*, 46(1), 90–105. Vo, X. V., & Ellis, C. (2018). International financial integration: Stock return linkages and volatility transmission between Vietnam and advanced countries. *Emerging Markets Review*, 36, 19–27. <https://doi.org/10.1016/j.ememar.2018.03.007>
- Wang, G. J., Xie, C., Lin, M., & Stanley, H. E. (2017). Stock market contagion during the global financial crisis: A multiscale approach. *Finance Research Letters*, 22, 163–168.
- Yadav, M. P., Bhatia, S., Singh, N., & Islam, T. (2022). Financial and energy exchange traded funds futures: An evidence of spillover and portfolio hedging. *Annals of Operation Research*. <http://doi.org/10.1007/s10479-022-04538-1>
- Yadav, M. P., Vasakarla, V., & Arora, M. (2020). Volatility spillover: Equity markets to commodity markets. *SCMS Journal of Management*, 17(3), 103–113.
- Yadav, N., Tandon, P., Tripathi, R., & Shastri, R. K. (2021). A dynamic relationship between crude oil price and Indian equity market: An empirical study with special reference to Indian benchmark index Sensex. *Benchmarking: An International Journal*, 28(2), 582–599.
- Yasir, M., & Önder, A. Ö. (2022). Time-varying herding spillover for emerging countries: evidence from BRIC countries and Turkey. *Review of Behavioral Finance*, advance online publication. <https://doi.org/10.1108/RBF-10-2021-0218>
- Yilmaz, K. (2010). Return and volatility spillovers among the East Asian equity markets. *Journal of Asian Economics*, 21(3), 304–313.
- Yousaf, M., Zahir, S., Riaz, M., Hussain, S. M., & Shah, K. (2020). Statistical analysis of forecasting COVID-19 for upcoming month in Pakistan. *Chaos, Solitons & Fractals*, 138, 109926.
- Yu, I. W., Fung, K. P., & Tam, C. S. (2010). Assessing financial market integration in Asia–equity markets. *Journal of Banking & Finance*, 34(12), 2874–2885.
- Zhang, W., Zhuang, X., & Wu, D. (2020). Spatial connectedness of volatility spillovers in G20 stock markets: Based on block models analysis. *Finance Research Letters*, 34, 101274. <https://doi.org/10.1016/j.frl.2019.08.022>
- Zhao, L., Saydaliev, H. B., & Iqbal, S. (2022). Energy financing, Covid-19 repercussions and climate change: Implications for emerging economies. *Climate Change Economics*, 13(3), 2240003. <https://doi.org/10.1142/S2010007822400036>

About the Authors



Nikhil Yadav is working as assistant professor—Senior Grade in Jaypee Business School, Jaypee Institute of Technology. He has completed PhD in finance from Motilal Nehru National Institute of Technology Allahabad & MBA (Finance) from UPTU Lucknow. He has 4 years of experience in the Mutual fund industry and equity research. Before joining Jaypee Business School, he has worked as assistant professor (finance) at Chandigarh University, Mohali Punjab. He is also an active reviewer of reputed journals of international publications indexed in Scopus, web of science and ABDC listed. He can be reached at nikhilsinghyadav87@gmail.com



Anurag Bhadur Singh has 13 years of teaching experience in the area of accounting and finance. He is an assistant professor (Sr. Grade) at XIM University, (XIMB) Bhubaneswar. He also served as faculty member at National Institute of Financial Management (Ministry of Finance, Government of India), Faridabad. He is trainer in the area of Procurement, General Financial Rules, Government Accounting & Budgeting and Government e-marketplace (GeM). He acquired D.Phil. in commerce from University of Allahabad (A Central University). His research interests span over financial markets and banking. He can be reached at mailtoanuragsingh.6324@gmail.com



Priyanka Tandon is working senior lecturer (finance) at Regenesys Business School-India Office. Prior to this, she served as Assistant Professor (Finance) in IIIT, Noida and as research associate in National Institute of Financial Management (MoF, GoI). She acquired her Doctor of Philosophy (Management) from Motilal Nehru National Institute of Technology, Allahabad (India), Master of Business Administration (Finance), Master of Commerce. She is also actively involved in research publication and has in her credit 15 publications till date. She is also working as reviewer for top ranked journals. She is among the top reviewers by Web of Science (Clarivate). She can be reached at neha.tandon886@gmail.com